Trunov Vladimir

Telegram | trunov.vv@phystech.edu | +7 (926) 692-03-22

Education

Moscow institute of physics and technology, Physics and Computer Science 2021 – Present

- **Bachelor's degree thesis:** Development of a domain-specific language for mapping nodes of the abstract syntactic tree of the SystemVerilog hardware description language
- **Master's degree thesis:** Development of a transparent data encryption module for PostgreSQL

Experience

C++ Software Engineer, Compilers Department, ISP RAS July 2022 – Dec 2023

Clang-Tidy/Clang Static Analyzer July 2022 - June 2023

Developed C/C++ source code vulnerability analysis tools based on Clang-Tidy and Clang Static Analyzer. Implemented a set of static checks according to AUTOSAR C++ 14 and Misra-C standards.

LLVM Backend July 2023 - Dec 2023

I wrote patches to the Clang compiler for the LLVM Backend extension to support code generation for a new processor based on the PowerPC architecture. I also added support from third-party libraries and tools such as Address Sanitizer and OpenMP.

Software Engineer, Yandex Jan 2024 – March 2025

I worked in the Yandex.Market Search and data preparation infrastructure department. My main language was C++, sometimes I had to use Python. I was engaged in the following tasks:

- Optimized the storage scheme in the database for fast search and low access timings.
- I rewrote the legacy big-data service, increasing its performance by **2-3 times**.
- I have written a new microservice that increases the fault tolerance of the system.

Research Software Engineer, Kaspersky March 2025 – Dec 2025

I worked in the department of testing and performance analysis. My responsibilities included:

- Profiling of Linux products using Linux Perf and various eBPF scripts.
- Analysis and optimization of product architecture.
- Development of internal infrastructure for automation of testing and profiling.

Software Engineer, Arenadata January 2026 – Present

My team is developing new open source storage engine OrioleDB for PostgreSQL. My achievements as a developer:

- Implemented DDL functionalities from the vanilla PostgreSQL engine.
- Profiled using Linux-based profilers. I had been involved in researching performance issues, and in this particular case, we achieved a **40x** decrease in disk usage.
- Developed fault injection crash-testing pipeline. That test helped to find and fix **10+** major bugs with concurrency and consistency.

Technologies

Languages: C/C++, Python, Go
Tools: Linux, clang/gcc, make, CMake, gdb/lldb, git, Linux Perf, Valgrind, eBPF, gRPC, Claude/Codex